

SkinSmith

A constraint-aware generative Agent for
game weapon skin design

Registered topic: Latent Vision → Image Generation

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**The output is judged on the mapped
asset—not on the source image alone.**

A good source image can still fail as a deployable texture

02

UV atlases split, rotate, and repack the visible surface. This creates failures that are invisible in the original artwork.

- **Fragmentation**

Motifs can disappear or be split across unrelated islands.

- **View dependence**

A design can read on the side but collapse in the narrow top view.

- **Deployment constraints**

Seams, paintable regions, resolution, and export format must all pass.



Source and mapped views are related evidence, but they are not interchangeable.

Three checkpoints separate interpretation, direction, and asset choice

03

1 Confirm the visual world

Choose a registered weapon and enter one keyword such as dragon. The Agent expands subjects, symbols, materials, palette, atmosphere, and forbidden content.

2 Select an art direction

Review 3–4 materially different textual directions. This resolves how the theme should be depicted before image generation.

3 Select a mapped artwork

Review 3–4 indivisible cards containing source + left/right/top views. Formal export cannot start before the human choice.

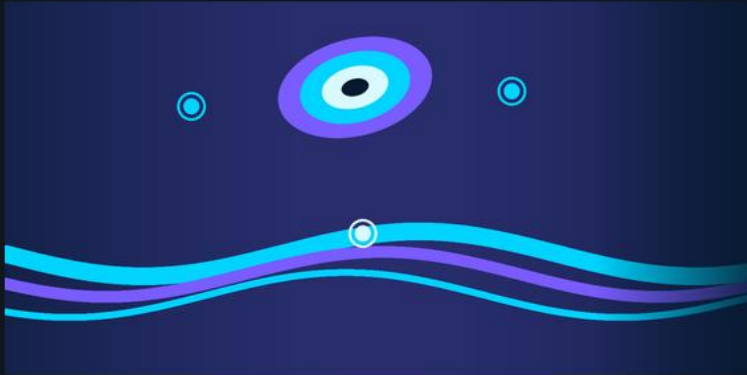
Model review can recommend; the user remains authoritative.

Weapon-space composition preserves intent before UV packing

04

1. Weapon-space design

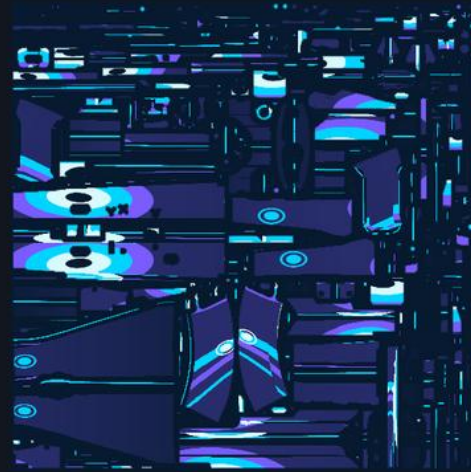
Coherent composition before UV packing



>

2. UV atlas storage

Fragments may look unrelated in 2D



>

3. 3D reconstruction

The model restores the intended whole



Route A

Pattern-first tileable baseline

Route B

Continuous weapon-space composition

Route C

One bounded render-conditioned correction

The UV atlas may look fragmented; the reconstructed weapon is the unit of evaluation.

Route B improved both seam feasibility and mapped-view quality

05

99.77%

mean asset-seam reduction

+0.060

mean multi-view improvement

4 / 4

paired candidates improved both

Mean seam error
0.26997 → 0.00063

Mean view score
0.76898 → 0.82911



ROUTE A · pattern-first baseline



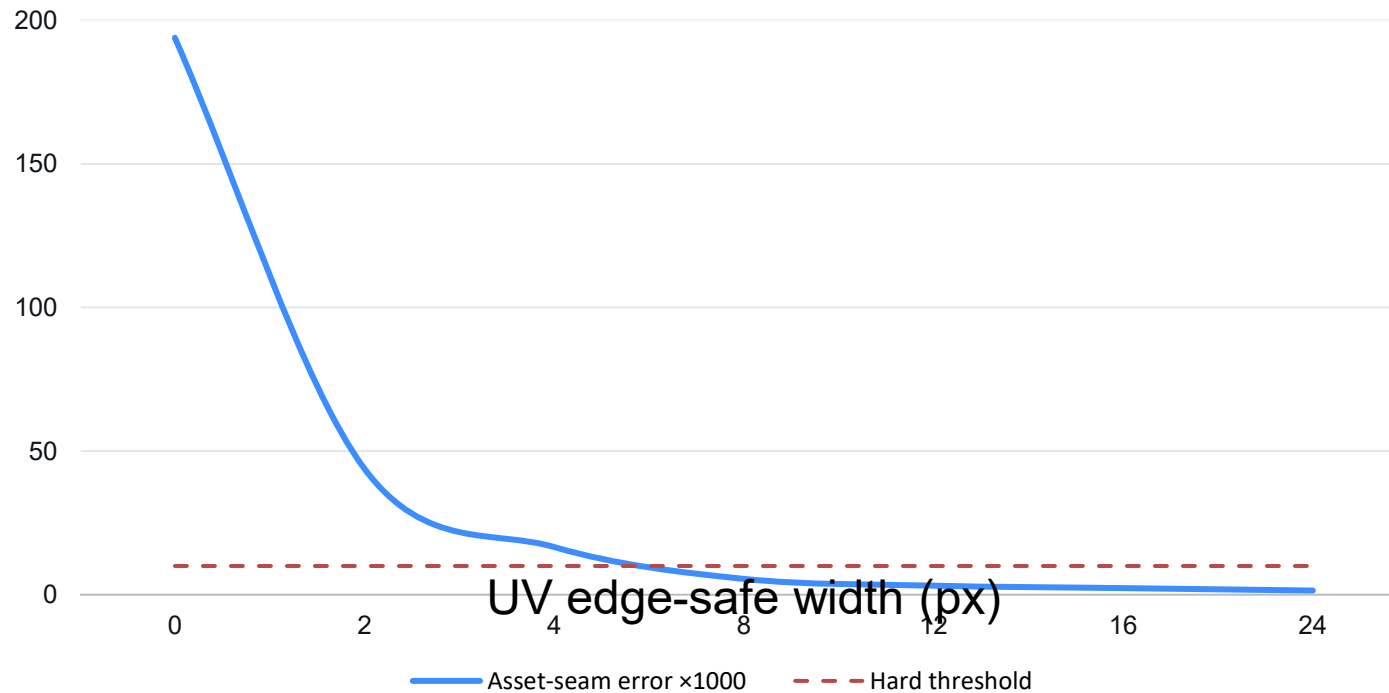
ROUTE B · weapon-space composition

Formal paired experiment, $n = 4$. Lower asset-seam error and higher multi-view score are better.

A weighted score selected an invalid candidate

06

The system treats feasibility as a gate before preference. Otherwise, a small visual gain can hide a deployment failure.



4 px

weighted choice · infeasible

8 px

constraint-first · feasible

24 px minimized seam error but failed the 95% multi-view-retention requirement.

The live Agent reused the exact human-selected source

07

dragon → Treasured Relic → artwork_04 → formal 2048 PNG/TGA export



Selected source artwork



0.00078

asset-seam
error

0.8166

multi-view score

0.8291

total score

The source hash and choice were preserved; the formal executor did not silently substitute a different image.

Bounded refinement can safely decide not to change the result

08

ROUND 0

0.8291

Accepted Route B
baseline

ROUND 1

0.8072

Local correction reduced
quality

DECISION

ROLL BACK

Score change -0.02196
Required improvement
 $+0.01000$

The contribution is not automatic self-improvement. It is observable diagnosis, bounded action, locality checking, and safe rejection.

The Agent contract transfers; asset parameters do not

09



AK-47 is the complete end-to-end case.
M4A4 is transfer evidence—not an equivalent production target.

Reusable across assets

● Agent state and checkpoints

Theme, direction, artwork, execution, and recovery phases.

● Generation and evaluation interfaces

Backends and evidence logging use shared contracts.

Must be recalibrated

OBJ/UV binding · seam graph · cameras · masks · edge width · thresholds

CONCLUSION

Evaluate the reconstructed asset, not the generated rectangle.

- 1 Generate in an asset-aware representation.
- 2 Keep source and mapped views in one human decision.
- 3 Gate feasibility before preference and retain rollback.

github.com/DID-FIELD/SDSC6002-SkinSmith

Group ownership

YUAN Ye
Project lead & integration

CHEN Yuhong
Creative planning & literature

Ben Tangjie
Generation & reproducibility

Wang Zhengye
Assets, UV & rendering

WANG Lihui
Evaluation & deliverable QA

Questions and discussion